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Deep Learning Project Report On **Deep Learning-Based Glacial Lake Segmentation Using Multi-Architecture Models**

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1. Introduction

Glaciers around the world are retreating at an accelerated rate due to rising global temperatures, with significant impacts observed in the Himalayan region. One of the most visible consequences of this retreat is the formation and expansion of glacial lakes. These lakes are formed from meltwater and often accumulate behind unstable natural dams such as moraines or ice barriers. As their number and size increase, they become important indicators of cryospheric change and environmental instability.

However, monitoring glacial lakes presents a significant challenge. Traditional mapping methods rely on manual interpretation of satellite imagery, which is time-consuming, labor-intensive, and prone to subjectivity. These limitations make it difficult to perform large-scale and frequent monitoring, especially in remote and rugged mountainous regions. Additionally, variations in terrain, lighting conditions, and spectral characteristics further complicate accurate identification of glacial lakes.

To address these challenges, automated approaches based on deep learning have emerged as a promising solution. Semantic segmentation models can perform pixel-level classification, enabling precise extraction of glacial lake boundaries from satellite imagery. Such automated methods not only improve accuracy but also significantly reduce processing time, making large-scale monitoring feasible. Therefore, developing efficient and reliable deep learning-based techniques for glacial lake segmentation is essential for advancing environmental monitoring and supporting future hazard assessment efforts.

2. Literature Review

2.1 Previous Work on Glacial Lake Mapping

Glacial lake mapping has traditionally relied on remote sensing techniques combined with manual or semi-automated approaches. Geographic Information System (GIS)-based methods and manual delineation using satellite imagery have been widely used to identify lake boundaries. However, these approaches are labor intensive and suitable only for limited spatial and temporal analysis.

To improve efficiency, researchers developed automated methods based on spectral indices and thresholding techniques using optical satellite data. These methods often require additional inputs such as Digital Elevation Models (DEM) and involve extensive post-processing to refine results. Despite their usefulness, such approaches struggle with complex terrain, shadows, and varying spectral characteristics of glacial lakes.

2.2 Use of Deep Learning in Segmentation

With the advancement of artificial intelligence, deep learning has become a powerful tool for glacial lake mapping. Convolutional Neural Networks (CNNs), particularly architectures like

U-Net and DeepLab, have been widely adopted for semantic segmentation tasks due to their ability to learn spatial and contextual features directly from data.

Recent studies have demonstrated high accuracy in glacial lake detection using deep learning models. For example, U-Net-based approaches have achieved strong performance in extracting lake boundaries from satellite imagery, effectively handling low contrast and complex backgrounds. Additionally, advanced models such as Generative Adversarial Networks (GANs) have been explored to further improve segmentation performance by capturing multi-level features and reducing dependency on handcrafted inputs. These developments highlight the growing role of deep learning in automating large-scale geospatial analysis.

2.3 Gaps in Existing Methods

Despite significant progress, several challenges remain in glacial lake mapping. Traditional methods rely heavily on handcrafted features and auxiliary data, limiting their adaptability and scalability. Even deep learning approaches face limitations such as high computational requirements, dependence on large annotated datasets, and reduced performance in complex environments like mountainous regions with shadows, snow, and cloud cover.

Furthermore, many existing studies focus on single-model approaches or limited datasets, making it difficult to generalize results across diverse geographical regions. There is also a need for comparative studies that evaluate multiple architectures under the same conditions. Addressing these gaps is essential for developing robust, scalable, and accurate solutions for glacial lake segmentation.

3. Dataset Description

This study utilizes the GLID (Glacial Lake Inventory Dataset), a large-scale annotated dataset specifically designed for glacial lake mapping using remote sensing imagery. The dataset provides diverse samples collected from multiple satellite platforms, making it suitable for training and evaluating deep learning models under varying environmental and imaging conditions.

The GLID dataset consists of a total of 18,367 samples, each with an original image size of 512×512 pixels. Every sample includes a corresponding annotation mask for supervised learning. The dataset is divided into training (16,000 samples) and validation (2,367 samples) sets, ensuring proper evaluation of model performance.

The dataset includes four types of glacial lakes: supraglacial lakes, proglacial lakes, ice-marginal lakes, and unconnected glacial lakes. These categories represent different physical characteristics and formation mechanisms, increasing the complexity and robustness of the dataset.

Annotations are provided as pixel-wise segmentation masks, where glacial lake regions are labelled with a value of 255 and the background is labelled as 0. Additionally, the dataset includes shapefiles with geographic information in the WGS 84 coordinate reference system, enabling geospatial analysis and visualization.

4. Methodology

The proposed approach focuses on automated glacial lake segmentation using deep learning-based semantic segmentation models. The methodology consists of a structured pipeline that includes data preprocessing, model training, and performance evaluation. Three state-of-the-art architectures U-Net, DeepLabV3, and Swin Transformer are implemented and compared to analyze their effectiveness in extracting glacial lake boundaries from satellite imagery.

4.1 Overall Pipeline

The overall workflow of the proposed system begins with input satellite images from the GLID dataset. These images undergo preprocessing to ensure uniformity in size and format. The processed images are then fed into deep learning models that perform pixel-wise classification to distinguish glacial lakes from background regions.

Each model generates a segmentation mask representing predicted lake regions. These predicted outputs are compared with ground truth annotation masks to evaluate performance. The evaluation is carried out using standard metrics such as Intersection over Union (IoU) and accuracy. The entire pipeline ensures a systematic comparison of different architectures under the same experimental conditions.

4.2 Data Preprocessing

Data preprocessing is a crucial step to ensure efficient model training and improved performance. The original GLID dataset consists of images with a resolution of 512×512 pixels. To reduce computational complexity and training time, all images were resized to 224×224 pixels. Corresponding annotation masks were resized using appropriate interpolation methods while preserving class labels (255 for glacial lakes and 0 for background).

In addition to resizing, input images were normalized to bring pixel values into a consistent range, which helps stabilize training and accelerate convergence. The dataset was divided into training and validation sets as provided in the GLID dataset (16,000 training samples and 2,367 validation samples). This split ensures unbiased evaluation of model performance.

4.3 Model Implementation

Three deep learning models were implemented for semantic segmentation: U-Net, DeepLabV3, and Swin Transformer. These models were selected to represent different architectural approaches, including convolutional neural networks and transformer-based methods.

4.3.1 U-Net

U-Net is a widely used convolutional neural network designed for image segmentation tasks. It follows an encoder–decoder structure. The encoder progressively extracts high-level features through convolution and pooling operations, reducing the spatial resolution while increasing feature depth. The decoder then reconstructs the segmentation map through upsampling operations.

A key feature of U-Net is the use of skip connections, which transfer feature maps from the encoder to the decoder. These connections help preserve spatial information and improve the accuracy of boundary detection. U-Net is particularly effective for capturing fine details and serves as a strong baseline model for segmentation tasks.

4.3.2 DeepLabV3

DeepLabV3 is an advanced segmentation model that improves upon traditional CNN-based approaches by incorporating atrous (dilated) convolutions. These convolutions allow the model to expand its receptive field without increasing the number of parameters, enabling better capture of contextual information.

The model also includes an Atrous Spatial Pyramid Pooling (ASPP) module, which processes feature maps at multiple scales using parallel convolutional layers with different dilation rates. This multi-scale feature extraction enables DeepLabV3 to handle objects of varying sizes and improves segmentation accuracy, especially in complex environments such as mountainous terrains.

4.3.3 Swin Transformer

Swin Transformer is a transformer-based architecture designed for computer vision tasks. Unlike convolutional models, it uses self-attention mechanisms to capture relationships between different regions of an image. The model divides the input image into patches and processes them using a hierarchical structure.

A unique feature of Swin Transformer is the shifted window attention mechanism, which allows interaction between neighboring windows and enhances contextual understanding. This enables the model to capture both local and global dependencies efficiently. As a result, Swin

Transformer is well-suited for segmenting glacial lakes in complex and heterogeneous landscapes.

4.5 Training Configuration

All models were trained under consistent conditions to ensure a fair comparison. The input image size was fixed at 224×224 pixels. A batch size of 8 was used during training to balance memory usage and computational efficiency. Each model was trained for 50 epochs to allow sufficient learning and convergence.

The training process involved minimizing a segmentation loss function, and performance was monitored on the validation dataset after each epoch. Training and validation loss, accuracy, and IoU were recorded to analyze model performance and convergence behavior.

To analyze the impact of hyperparameters, experiments were conducted using two different learning rates. All models were trained under identical conditions, and performance was evaluated using loss, accuracy, IoU, and Dice coefficient.

4.6 Evaluation Strategy

The performance of each model was evaluated using standard segmentation metrics. Intersection over Union (IoU) was used as the primary metric to assess the overlap between predicted and ground truth masks. Accuracy was also considered to measure overall pixel-wise classification performance.

Both quantitative metrics and qualitative visual comparisons (input image, ground truth, and predicted mask) were used to evaluate the effectiveness of each model. This comprehensive evaluation ensures a robust comparison of different architectures for glacial lake segmentation.

5. Model Architecture:

To perform accurate glacial lake segmentation, three state-of-the-art deep learning architectures were implemented: U-Net, DeepLabV3, and Swin Transformer. These models represent different approaches to semantic segmentation, including convolution-based and transformer-based methods.

5.1 U-Net Architecture:

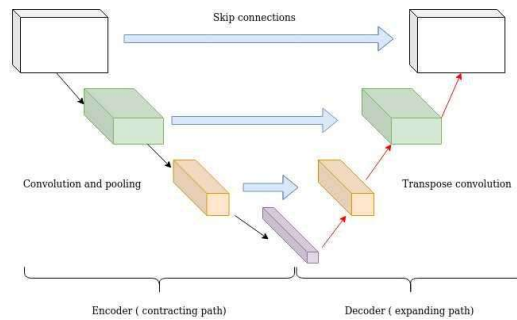


Fig. Unet

U-Net is a convolutional neural network specifically designed for image segmentation tasks. It follows a symmetric encoder-decoder architecture:

Key Components:

- **Encoder (Contracting Path):**
The encoder consists of repeated convolutional layers followed by ReLU activation and max-pooling operations. It progressively reduces spatial dimensions while extracting high-level features.
- **Decoder (Expanding Path):**
The decoder performs upsampling using transposed convolutions to restore spatial resolution and generate the segmentation map.
- **Skip Connections:**
Feature maps from the encoder are directly concatenated with corresponding decoder layers. This helps retain fine-grained spatial information lost during downsampling.

Advantages:

- Excellent at capturing fine details and boundaries.
- Works well with relatively small datasets.
- Strong baseline for segmentation tasks.

Limitation:

- Limited ability to capture global contextual information.

5.2 DeepLabV3 Architecture:

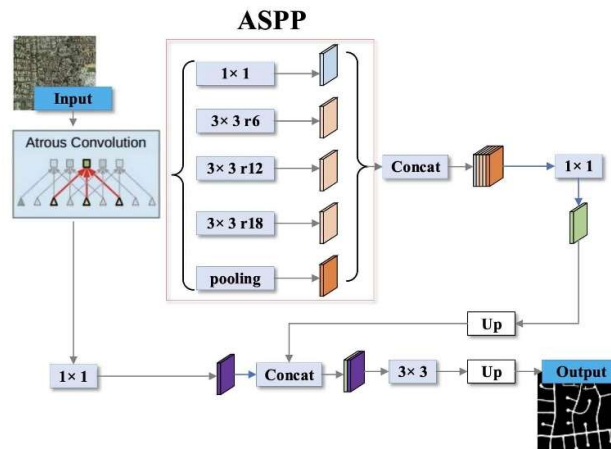


Fig. DeepLab

DeepLabV3 improves segmentation performance by incorporating multi-scale context and dilated convolutions.

Key Components:

- **Backbone Network:**
A CNN (e.g., ResNet) is used to extract feature maps from the input image.
- **Atrous (Dilated) Convolutions:**
These expand the receptive field without reducing resolution, allowing the model to capture broader contextual information.
- **Atrous Spatial Pyramid Pooling (ASPP):**
ASPP applies multiple parallel atrous convolutions with different dilation rates, enabling multi-scale feature extraction.

Advantages:

- Strong multi-scale feature representation.
- Better handling of varying object sizes.
- Improved boundary segmentation compared to standard CNNs.

Limitation:

- Higher computational complexity than U-Net.

5.3 Swin Transformer Architecture:

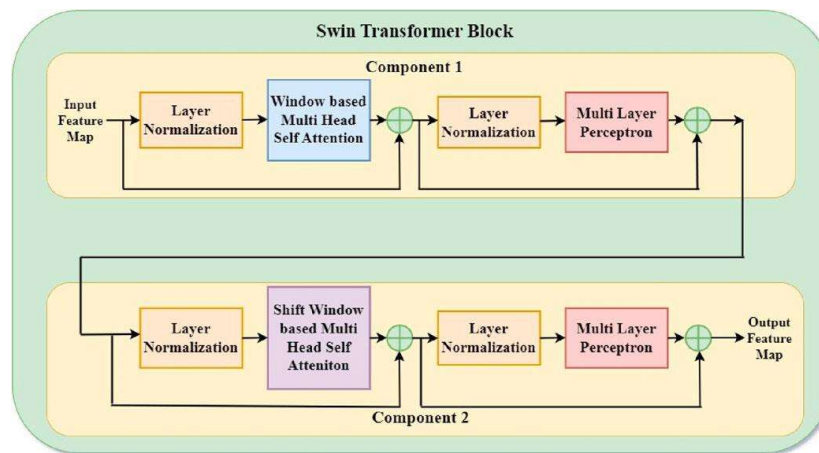


Fig. Swin Transformer

Swin Transformer is a transformer-based architecture adapted for vision tasks, designed to overcome limitations of traditional CNNs.

Key Components:

- **Patch Partitioning:**
The input image is divided into non-overlapping patches, which are treated as tokens.
- **Hierarchical Structure:**
The model processes features at multiple scales, similar to CNNs, enabling efficient representation learning.
- **Shifted Window Self-Attention:**
Instead of global attention, Swin Transformer computes attention within local windows and shifts them between layers to allow cross-window interaction.

Advantages:

- Captures long-range dependencies.
- Strong global contextual understanding.
- Effective in complex and heterogeneous scenes.

Limitation:

- Requires more data and computational resources.

6. Training Details

The segmentation models were trained under a consistent experimental setup to ensure fair comparison across architectures. The training process was designed to achieve stable convergence while maintaining computational efficiency.

The models were trained using two different learning rates (0.001, 0.0001) to study their impact on performance.

6.1 Loss Function

A segmentation loss function was used to guide the learning process by measuring the difference between predicted masks and ground truth annotations. In this study, **Binary Cross-Entropy (BCE) Loss** was used, as the task involves binary classification (glacial lake vs background).

The loss function penalizes incorrect pixel-wise predictions and helps the model learn accurate segmentation boundaries. Minimizing this loss enables the models to improve their ability to distinguish between lake and non-lake regions.

6.2 Optimizer

The **Adam optimizer** was employed for training all models due to its efficiency and adaptive learning rate mechanism. Adam combines the advantages of momentum and RMSProp, allowing faster convergence and better handling of sparse gradients.

It dynamically adjusts learning rates for each parameter, making it suitable for deep learning models with complex architectures such as U-Net, DeepLabV3, and Swin Transformer.

6.3 Training Configuration

- **Epochs:** 50
- **Batch Size:** 8
- **Input Image Size:** 224×224

All models were trained for 50 epochs, allowing sufficient time for convergence. A batch size of 8 was selected to balance memory constraints and training stability.

During training, both training and validation performance were monitored using loss, accuracy, and IoU metrics. This helped in analyzing convergence behavior and ensuring that the models generalize well without overfitting.

7. Evaluation Metrics

To assess the performance of the segmentation models, standard metrics for pixel-wise classification were used, including Intersection over Union (IoU), Dice Coefficient, and Accuracy. These metrics provide a comprehensive evaluation of how well the predicted segmentation masks match the ground truth.

7.1 Intersection over Union (IoU)

Intersection over Union (IoU), also known as the Jaccard Index, measures the overlap between the predicted segmentation and the ground truth.

$$\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$$

Where:

- **TP (True Positive):** correctly predicted lake pixels
- **FP (False Positive):** background pixels incorrectly predicted as lake
- **FN (False Negative):** lake pixels missed by the model

IoU ranges from 0 to 1, where a higher value indicates better segmentation performance.

7.2 Dice Coefficient

The Dice Coefficient is another similarity measure that evaluates the overlap between predicted and ground truth masks.

$$\text{Dice} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$$

It is closely related to IoU but gives more weight to correctly predicted pixels. Dice values also range from 0 to 1, with higher values indicating better performance.

7.3 Accuracy

Accuracy measures the proportion of correctly classified pixels over the total number of pixels.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Where:

- **TN (True Negative):** correctly predicted background pixels

While accuracy provides an overall measure, it may not fully reflect segmentation quality in imbalanced datasets.

7.4 Importance of IoU

Among all metrics, IoU is considered the most important for segmentation tasks because it directly evaluates the quality of overlap between predicted and actual regions. Unlike accuracy, IoU is not biased toward dominant classes (background), making it more reliable for datasets where the object of interest (glacial lakes) occupies a smaller portion of the image.

Therefore, IoU is used as the primary metric for comparing model performance in this study.

8. Results

The performance of the implemented models—U-Net, DeepLabV3, and Swin Transformer—was evaluated using quantitative metrics and visual analysis. The results include numerical comparisons as well as training behavior observed through graphs.

8.1 Quantitative Results

The table below summarizes the performance of all models on training and validation datasets:

Model	Training Loss	Validation Loss	Training Accuracy	Validation Accuracy	Training IoU	Validation IoU
Unet	0.051788	0.087095	0.975974	0.967641	0.032996	0.034828
DeepLab	0.03046	0.056704	0.975974	0.967641	0.051573	0.040537
Swin Transformer	0.073884	0.08658	0.975974	0.967641	0.021919	0.023208

Table 1: Model Comparison

Key Observations:

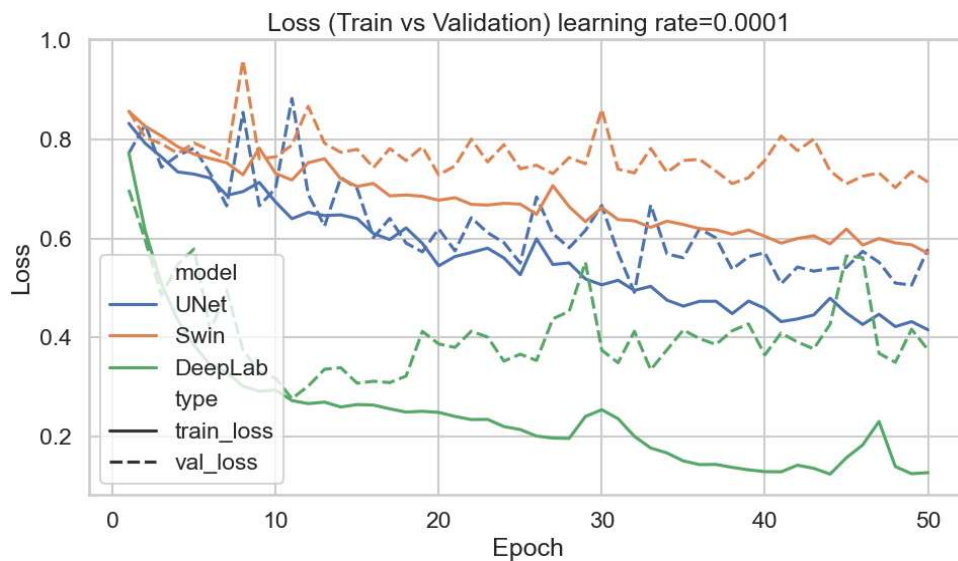
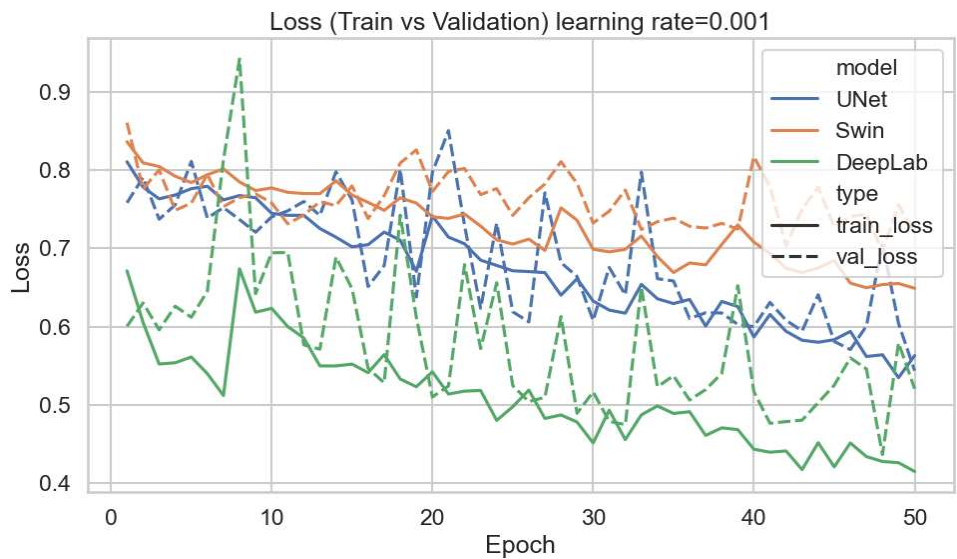
- DeepLabV3 achieves the lowest loss values, indicating better convergence.
- DeepLabV3 also achieves the highest IoU, making it the best-performing model overall.
- U-Net shows stable and competitive performance, serving as a strong baseline.
- Swin Transformer shows lower IoU, indicating room for improvement under current training settings.

- Accuracy is similar across all models, suggesting it is less effective for distinguishing performance in this task.

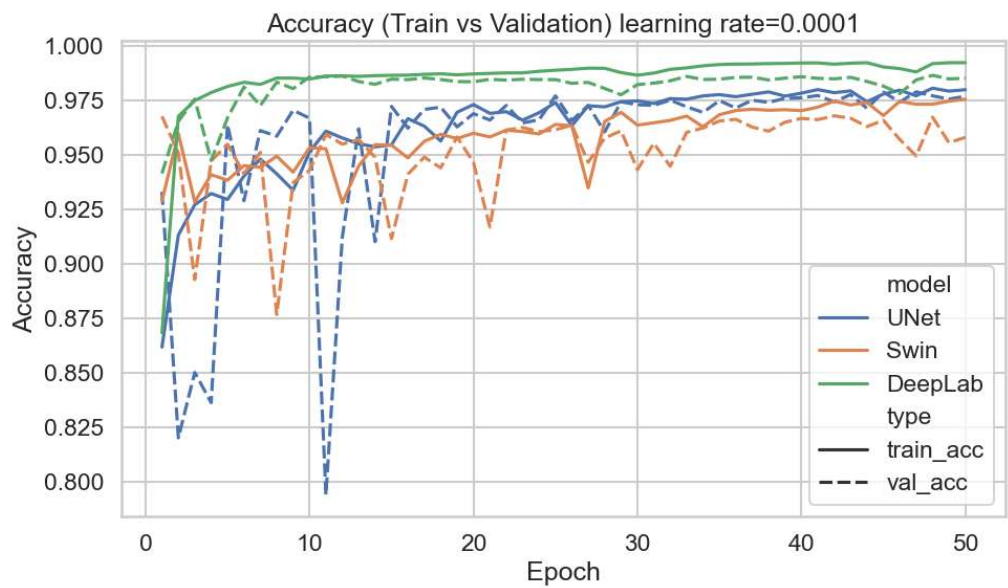
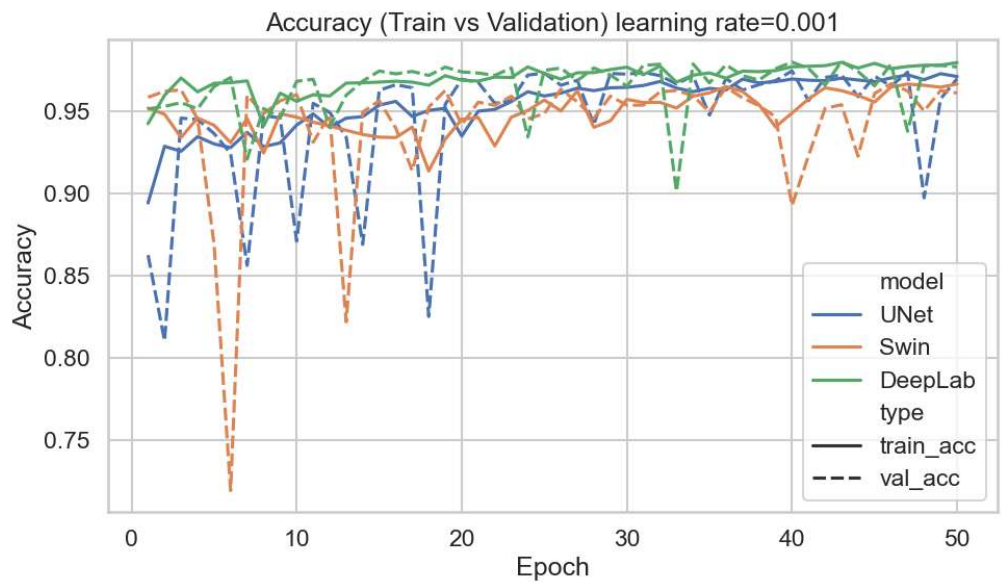
8.2 Graphical Analysis (Effect of Learning Rate):

The performance of the models was evaluated under different learning rates to analyze their impact on convergence and segmentation accuracy. The following graphs compare training and validation metrics across multiple learning rates.

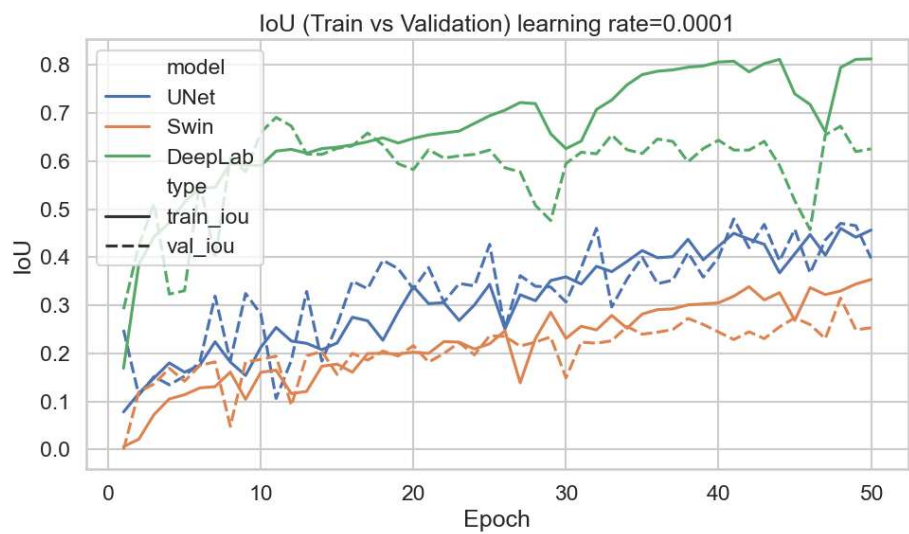
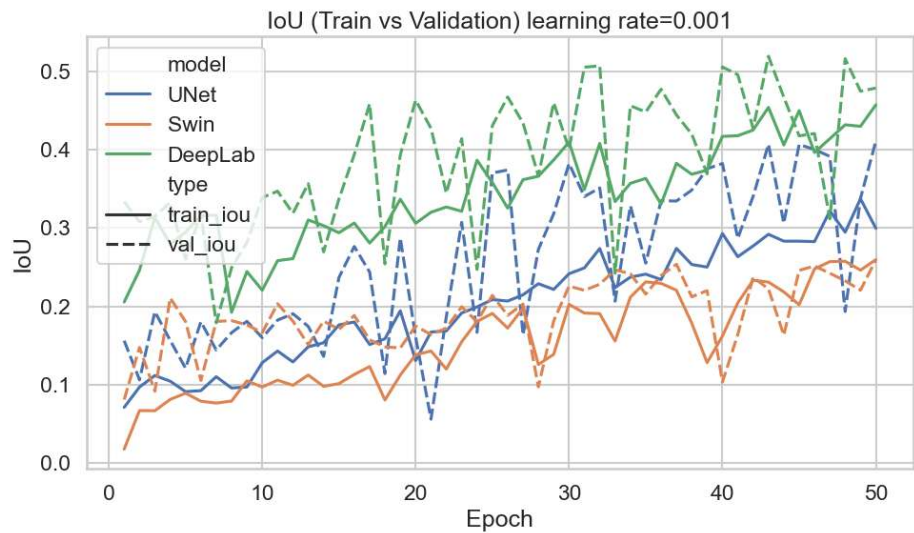
8.2.1 Loss Graph



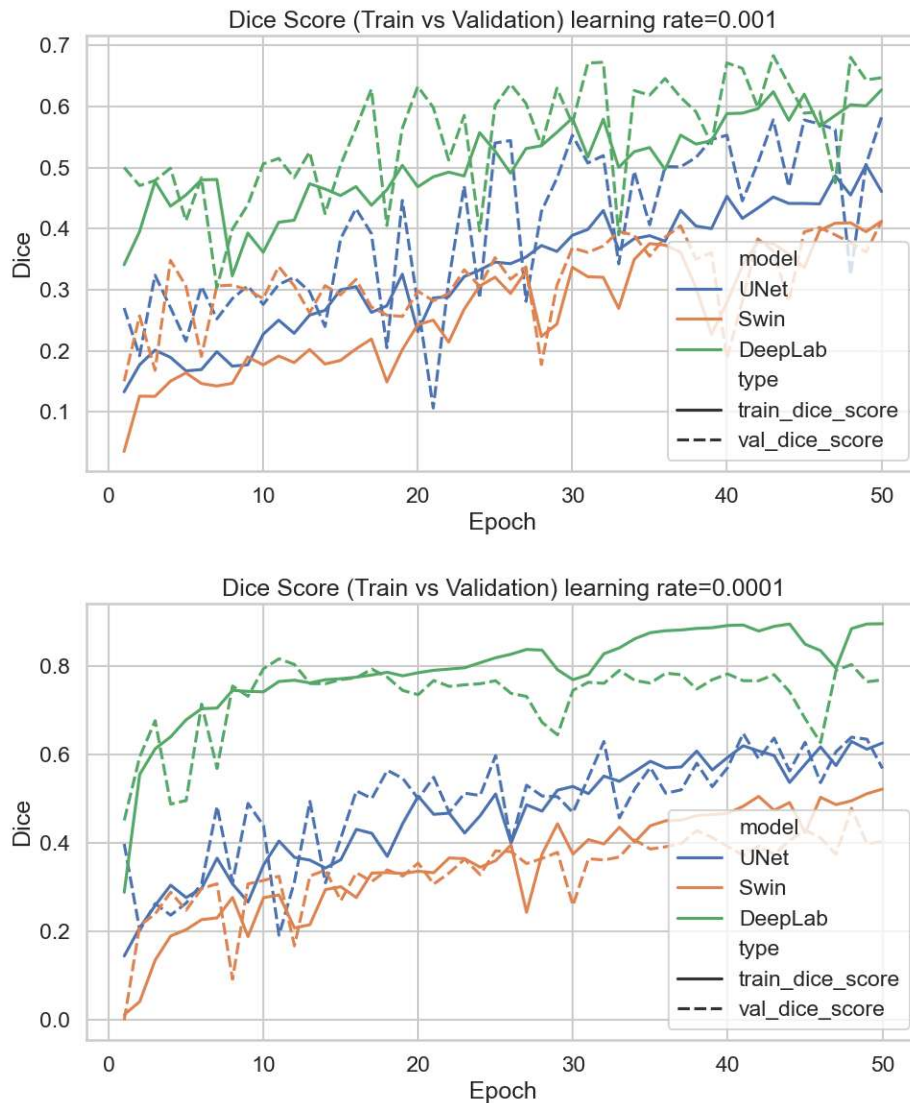
8.2.2 Accuracy Graph



8.3.3 IoU Graph



8.3.4 Dice Score



The results indicate that learning rate significantly impacts model performance. Higher learning rates lead to faster convergence but may introduce instability, while lower learning rates provide more stable but slower learning. The optimal learning rate achieves a balance between convergence speed and segmentation accuracy, resulting in improved IoU and Dice scores.

9. Model Comparison

The performance of U-Net, DeepLabV3, and Swin Transformer was compared using quantitative metrics and visual analysis. Among the three models, DeepLabV3 achieved the best overall performance, with the lowest loss values and highest IoU scores, indicating superior segmentation capability. Its use of atrous convolutions and multi-scale feature extraction enables better handling of varying lake sizes and complex backgrounds.

U-Net demonstrated stable and consistent performance, serving as a reliable baseline model. It effectively captured general lake structures and showed steady improvement during training. However, its ability to capture global contextual information is limited compared to more advanced architectures.

The Swin Transformer, while conceptually powerful due to its ability to model long-range dependencies, showed comparatively lower IoU under the current training setup. This suggests that transformer-based models may require more data, training time, or hyperparameter tuning to fully realize their potential.

Each model exhibits distinct strengths and limitations:

- **U-Net**
 - Strengths: Simple architecture, stable training, good boundary detection
 - Weaknesses: Limited global context understanding
- **DeepLabV3**
 - Strengths: Strong multi-scale feature extraction, best segmentation performance, faster convergence
 - Weaknesses: Higher computational complexity
- **Swin Transformer**
 - Strengths: Captures global context, effective for complex scenes
 - Weaknesses: Higher training requirements, lower performance under current configuration

9.1 Observations from Results

- All models show consistent learning behavior, as indicated by decreasing loss curves.
- Accuracy remains high across models, suggesting that accuracy alone is not sufficient for evaluating segmentation performance.
- IoU provides a more meaningful comparison, where DeepLabV3 clearly outperforms other models.
- The small gap between training and validation metrics indicates good generalization and minimal overfitting.

- Visual results confirm that models can successfully detect glacial lakes, though minor errors occur in small or complex regions.

10. Visual Results

To qualitatively evaluate model performance, visual comparisons were made between the input satellite images, corresponding ground truth masks, and predicted segmentation outputs. The input images represent raw satellite data, while the ground truth masks indicate the actual glacial lake regions annotated in the dataset. The predicted masks generated by the models highlight the detected glacial lake areas.

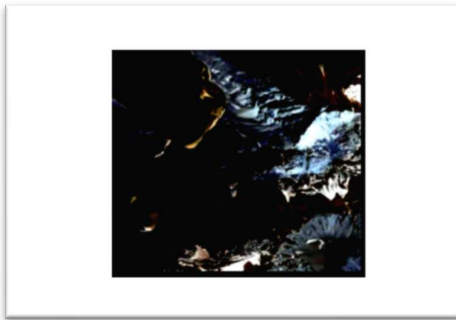


Fig. Input Image



Fig. Label



Fig. Unet Prediction



Fig. DeepLab Prediction



Fig. Swin Transformer Prediction

The visual results demonstrate that all models are capable of identifying glacial lakes from complex backgrounds. The predicted masks show strong agreement with the ground truth, indicating accurate segmentation performance.

DeepLabV3 produces more refined and continuous lake boundaries, closely matching the annotated regions. U-Net captures the general shape of lakes effectively but may show slight

boundary inaccuracies in complex regions. Swin Transformer demonstrates the ability to detect broader contextual patterns but may miss smaller or less distinct lake areas under the current training setup.

Some minor discrepancies are observed in challenging conditions such as shadowed regions, mixed ice-water areas, and small fragmented lakes. Despite these challenges, the models successfully capture the overall distribution and structure of glacial lakes.

11. Conclusion

This study presents a deep learning–based approach for automated glacial lake segmentation using the GLID dataset. Three models—U-Net, DeepLabV3, and Swin Transformer were implemented and evaluated using standard segmentation metrics. The results demonstrate that all models are capable of accurately identifying glacial lakes from satellite imagery, with consistent learning behavior and good generalization performance.

Among the evaluated models, DeepLabV3 achieved the best overall performance, showing the lowest loss and highest IoU scores. Its ability to capture multi-scale contextual information enables more accurate and refined segmentation, particularly in complex terrains. U-Net provided stable and reliable baseline performance, while Swin Transformer showed potential for capturing global context but requires further optimization.

The key takeaway from this work is that deep learning enables accurate, scalable, and automated glacial lake mapping, significantly reducing dependence on manual methods. This study establishes a strong foundation for future applications in large-scale environmental monitoring and glacial hazard assessment.

12. Future Work

While the proposed approach demonstrates effective glacial lake segmentation, several improvements can be explored in future work. Model performance can be further enhanced by optimizing hyperparameters, increasing training epochs, and experimenting with advanced loss functions such as Dice loss or combined loss functions. Additionally, fine-tuning transformer-based models and using larger input resolutions may improve segmentation accuracy, especially for small and complex lake regions.

Another important direction is the incorporation of additional and more diverse datasets from different geographic regions and satellite platforms. This would improve model generalization and robustness under varying environmental conditions.

Finally, the segmentation framework can be extended to downstream applications such as Glacial Lake Outburst Flood (GLOF) risk prediction. By integrating segmented lake features (e.g., size, location, proximity to glaciers) with geospatial and environmental data, a

comprehensive predictive system can be developed to support early warning and disaster risk management.

13. References

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